Geometry-Aware Diffusion for Controllable Multi-View Aerial Generation

Na Yan, Siyu Zhu, Kaiji Hou, Xiulei Liu, and Qiang Tong(✉)

College of Computer Science, Beijing Information Science and Technology University, Beijing, China

Abstract. Multi-view image generation methods often suffer from geometric inconsistency and texture discontinuity due to limited 3D scene understanding. This paper presents View-Diff, a novel geometry-aware diffusion-based framework for controllable and high-quality multi-view image generation in aerial scenes. The key innovations comprise two aspects: 1) A Dual-Branch pipeline consists of two parts. The first is a viewpoint transformation branch that precisely transforms known regions. The second is a Spatial Guidance Module branch, which employs scene geometry features to guide the realistic completion of unknown regions and maintain spatial consistency across different viewpoints. 2) an Optimized Perceptual Module incorporates structure-preserving perceptual loss to strengthen feature preservation in known regions, thereby boosting overall coherence. Extensive experiments were conducted on multiple aerial multi-view datasets. View-Diff outperforms state-of-the-art baselines. It achieves notable gains in I2I, LPIPS, and FID on SUES-200 and University-1652. It also shows significant improvements in viewpoint accuracy and visual quality.

Keywords: Computer Vision · Deep Learning · Multi-View Image Generation · Viewpoint Control

1 Introduction

Multi-view image generation refers to the task of generating novel images of the same scene from different viewpoints based on a single-view image, thereby expanding the range of image viewpoints. This task is of great significance for downstream applications such as image object detection [1], 3D reconstruction [2], as it helps mitigate the challenges caused by viewpoint limitations and occlusions. However, in complex aerial scenes, existing methods struggle to ensure both geometric consistency and high-fidelity texture in the generated views. Moreover, most approaches lack the ability to precisely control the target viewpoint, leading to spatial misalignment and limited visual quality.

Early methods based on Variational Autoencoders (VAEs) [3] learned latent representations through variational inference, but often produced blurry and low-resolution outputs. 3D-based pipelines [4] attempt to reconstruct and render 3D scenes. however, these approaches typically rely on rigid assumptions and fail to

generalize in complex real-world environments. With the emergence of GANs [5], image realism improved. For instance, CR-GAN [6] introduced dual-path training, and VI-GAN [7] used view-invariant features for better viewpoint control. Nevertheless, such models often suffer from mode collapse, unstable training, and heavy dependency on annotated camera poses. Diffusion models, particularly Denoising Diffusion Probabilistic Models (DDPMs) [8], offer a more stable and interpretable alternative. They progressively transform noise into detailed images, outperforming GANs in terms of fidelity and convergence. However, viewpoint control remains challenging. Due to their stochastic nature, diffusion-based models often require paired source-target views for supervision, which limits their scalability. Recent efforts like GeNVS [9] explore implicit viewpoint alignment using category priors, but still suffer from geometric inconsistencies and limited cross-category generalization.

To address these challenges, we propose View-Diff, a geometry-aware and controllable diffusion-based framework for multi-view aerial image generation. Our method focuses on reducing geometric inconsistency and texture discontinuity while enabling precise viewpoint control. It employs a Dual-Branch Pipeline to disentangle viewpoint transformation and spatial completion guidance. In addition, it proposes an Optimized Perceptual Module to enhance structural preservation and perceptual quality.

In summary, the main contributions of this work are as follows:

- We propose View-Diff, a controllable multi-view image generation method that enables accurate viewpoint control through an explicit constraint mechanism. This mitigates the randomness often observed in existing methods and reduces the reliance on viewpoint-specific datasets, thereby significantly improving the model's generalization on generic datasets.
- We design a Dual-Branch Pipeline to control the transformation of known regions and guide the generation of unknown regions. This improves spatial consistency and increases viewpoint accuracy in the generated images across viewpoints.
- We introduce an Optimized Perceptual Module that enhances the model's ability to perceive and preserve structural information in known regions, thereby improving the overall quality of the generated novel-view images.
- Extensive experiments have been conducted on two public datasets, SUES-200 and University-1652. Our proposed method demonstrates its effectiveness and superiority by achieving improvements in I2I, LPIPS, and FID over current state-of-the-art methods.

2 Related work

2.1 Image Generation

Recent advances in deep learning have significantly advanced the field of image generation. Early methods like VAEs [3] offered stable training and theoretical

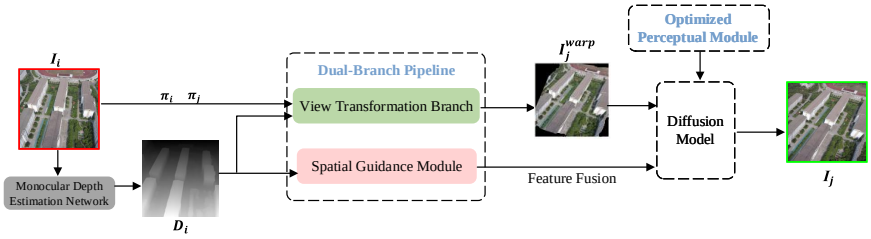

Fig. 1. Pipeline of the proposed View-Diff method.

clarity but often produced blurry images due to limited detail modeling. GANs introduced adversarial learning, greatly improving realism. Variants such as Conditional GANs [10], and CycleGANs [11] extended GANs to various conditioning scenarios, but suffered from training instability and mode collapse.

Denoising Diffusion Probabilistic Models (DDPMs) [8] emerged as a more stable alternative, offering strong theoretical grounding and high-quality results. Subsequent variants like DDIM [12] improved sampling efficiency. Diffusion models have since become a leading approach in image generation. However, when applied to complex scenarios such as aerial scenes, existing methods still face considerable challenges. These include geometric inconsistency, and texture discontinuity, primarily due to insufficient global spatial reasoning. These challenges hinder structural coherence and perceptual quality, necessitating a geometry-aware and viewpoint-controllable framework for aerial image generation.

2.2 Mult-view Image Generation

Early novel view generation methods relied on 3D reconstruction and rendering from new viewpoints. For instance, 3D-R2N2 [13] learn volumetric 3D representations from 2D inputs or latent codes, enabling coarse structural recovery. Cross-view generation approaches such as SelectionGAN [14], apply domain translation strategies using attention-based encoders to bridge viewpoint gaps. However, these methods often rely on large-scale or labeled datasets and lack the capability for precise viewpoint control.

Recent diffusion-based models such as GeNVS [9] demonstrate improved spatial coherence by introducing pose or geometric priors. Nevertheless, they typically focus on 3D-aware consistency rather than explicit and fine-grained 2D viewpoint transformation. Our approach is different in that targets fine-grained 2D view control with spatial guidance to improve structural and semantic alignment.

Moreover, although general-purpose generative models such as Crfill [15], MAT [16], BLD [17], and SPA-Former [18] are not specifically designed for novel view generation, they demonstrate strong performance in contextual modeling and visual fidelity. However, In aerial scenes featuring complex spatial layouts

and fine-grained semantic structures, these models often fail to preserve geometric consistency or produce continuous textures across transformed viewpoints. Specifically, their limited understanding of global scene geometry leads to issues such as broken textures, and inconsistent spatial alignment in the generated views.

To address these limitations, Our method explicitly integrates viewpoint control and spatial guidance, leading to more robust and consistent novel view generation.

3 View-Diff

Aerial scenes pose significant challenges for novel view generation due to their complex structures and rich semantics. Existing methods often struggle with precise viewpoint control and high-quality results without annotated data. To address this, we propose View-Diff, a controllable multi-view framework that improves viewpoint accuracy, structural consistency, and visual quality.

The pipeline is shown in Fig. 1. Given an input image I_i, a pretrained depth estimation network [19] generates a depth map D_i. Along with input and target camera parameters π_i and π_j, these feed into a Dual-Branch Pipeline. The view transformation branch uses depth and camera parameters to produce a warped image I_j^{warp} of known regions. The spatial guidance module extracts spatial features from D_i to assist structural modeling. Features from both branches pass to a pretrained diffusion model [20]. Finally, an Optimized Perceptual Module enhances the model's understanding of I_j^{warp} and guides plausible content generation in unknown regions, producing a novel-view image I_j aligned with the target viewpoint.

3.1 Task Definition

We first provide a brief definition of the multi-view image generation task. Suppose there exists a set of viewpoints $V = \{v_1, v_2, \ldots, v_d\}$, where each $v \in V$ corresponds to a specific view. Given an image I_i captured from viewpoint v_i, the task of multi-view generation is to generat an image I_j under a target viewpoint v_j. This is conditioned on the input image I_i, the source camera parameters π_i for v_i, and the target camera parameters π_j for v_j, where $v_j \in V$ and $j \neq i$.

3.2 Dual-Branch Pipeline

To tackle inaccurate viewpoint control and spatial inconsistency in novel view generation, we propose a Dual-Branch Pipeline. Existing methods often combine geometric transformation and spatial completion, causing blurred boundaries and texture breaks, especially in complex aerial scenes with large unknown regions. Our pipeline separates these tasks into two branches: the view transformation branch models geometric changes in known regions for accurate viewpoint fidelity, while the spatial guidance module provides structural priors to

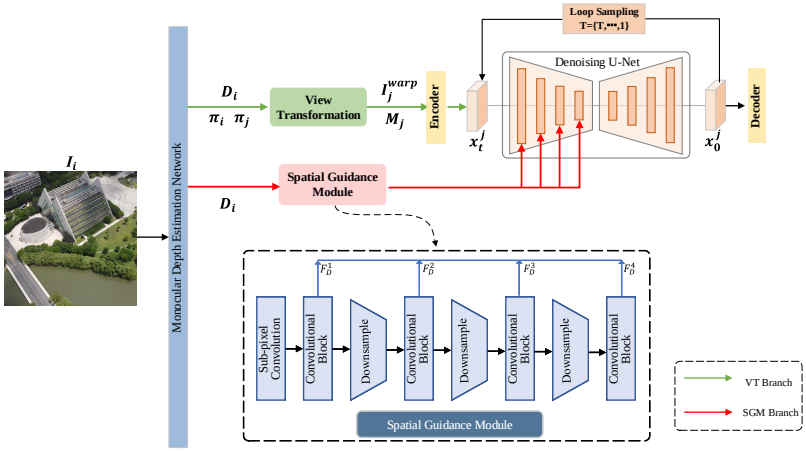

Fig. 2. Architecture of the Dual-Branch Pipeline. Here, the green branch represents View Transformation Branch(VT Branch), and the red branch denotes Spatial Guidance Module Branch(SGM Branch).

guide plausible generation in unknown areas. This division enables precise viewpoint control and spatial consistency, effectively reducing geometric and texture inconsistencies present in prior methods.

The architecture of the proposed Dual-Branch Pipeline is illustrated in Fig. 2. First, as in previous works [21], We use a pretrained monocular depth estimation network [19] to estimate the depth map D_i from the input image I_i under viewpoint v_i. This depth map is then passed to both branches.

$$D_i = \mathcal{F}_{\text{depth}}(I_i), \quad p_j = \pi_j \left(\pi_i^{-1}(p_i, D_i(p))\right) \tag{1}$$

where D_i is the depth map estimated from the input image I_i , p_i denotes a pixel coordinate in I_i, and p_j is the corresponding projected coordinate in the target view; $\mathcal{F}_{\text{depth}}$ represents a pretrained monocular depth estimation network.

In View Transformation Branch, depth-guided viewpoint transformation is performed. Each pixel is mapped to a 3D world coordinate using D_i and camera parameters π_i. then, projected into the 2D space under target camera parameters π_j, producing a preliminary target view image I_j^{warp} in known regions and a visibility mask M_j. Its calculation formula is defined as:

$$I_j^{\text{warp}}, M_j = \mathcal{F}_{\text{warp}}(I_i, D_i, \pi_i, \pi_j) \tag{2}$$

where $\mathcal{F}_{\text{warp}}$ denotes the warping function based on camera geometry; I_j^{warp} represents the warped known regions from the source view to the target view; M_j is the visibility mask indicating unknown regions in the target view.

Then I_j^{warp} and M_j are fed into the diffusion model, where the encoder extracts multi-scale features:

$$F_{enc}^k = \mathcal{F}_{\text{enc}}(I_j^{\text{warp}}, M_j), \quad k \in \{1, 2, 3, 4\} \tag{3}$$

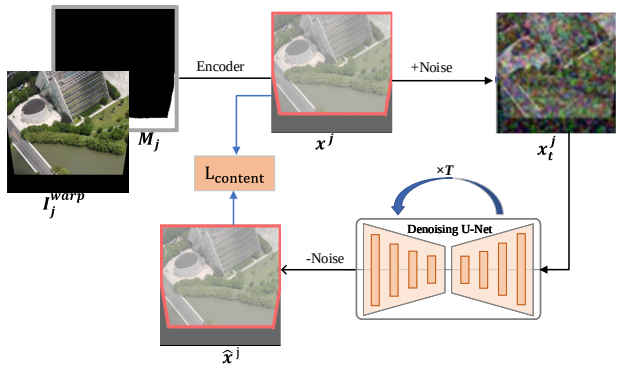

Fig. 3. Architecture of the Optimized Perceptual Module.

Where $\mathcal{F}_{\text{enc}}$ denotes the encoder operation, F_{enc}^k are the multi-scale features extracted at different levels of the U-Net.

Concurrently, in Spatial Guidance Module Branch takes the estimated depth map D_i as input to the Spatial Guidance Module(SGM), which rearranges it into a spatially structured feature representation for multi-scale extraction. The features are then processed through convolutional and downsampling layers to match the U-Net [22] encoder dimension.

$$F_D^k = \mathcal{F}_{\text{SGM}}(D_i), \quad k \in \{1, 2, 3, 4\} \tag{4}$$

where F_{SGM} refers to the Spatial Guidance Module which extracts multi-scale features F_D^k to guide the generation of unknown regions.

Finally, the spatial guidance features are integrated with the encoder outputs through element-wise addition at each corresponding scale, enhancing the geometric consistency of the encoded representation. The operation is defined as:

$$F^k = F_{\text{enc}}^k \oplus F_D^k, \quad k \in \{1, 2, 3, 4\} \tag{5}$$

Where $\oplus$ denotes element-wise addition. The resulting feature F^k combines both appearance information from known regions and structural information from the depth map.

3.3 Optimized Perceptual Module

To address the problem of texture discontinuity and insufficient global structural understanding in unknown regions, we propose an Optimized Perceptual Module (OPM). This module is designed to refine the reconstruction of known regions and enhance the understanding of semantic information and global structure in the known regions, thereby allowing the model to focus more of its generative capacity on the unknown regions.

The architecture of the proposed Optimized Perceptual Module is illustrated in Fig. 3. Given the warped image I_j^{warp} and its associated mask M_j, we first encode them into a low-dimensional latent space to obtain the feature representation x^j. This latent representation encodes the semantic and structural information of known regions, reducing computational cost while preserving essential features for generation.

During the diffusion process, the latent feature x^j is gradually perturbed by adding Gaussian noise over a sequence of timesteps t, eventually producing a noisy sample x_t^j. In the denoising process, the model attempts to recover the clean latent x^j by iteratively predicting and removing the noise component from x_t^j. Unlike conventional diffusion-based generative models that compute reconstruction loss over the entire image, our method restricts the loss computation only to the known regions. This selective supervision ensures that learning focuses on regions with reliable ground-truth information, preventing the model from overfitting to noise patterns in unknown regions.

Furthermore, to maintain semantic consistency across sampling steps, we introduce a lightweight latent calibration mechanism. Specifically, at each sampling step, we apply weak alignment between the current latent representation and the original noisy latent in known regions. This strategy reduces accumulated errors during denoising and preserves clean gradients in known areas, thereby mitigating interference in unknown regions and improving the coherence and plausibility of the generated results. Our optimized perceptual loss function is defined as :

$$\ell_{\text{content}} = \begin{cases} \mathbb{E}_{t,x_0,\epsilon} \left\| \epsilon - \epsilon_\theta(x_t, t) \right\|_2^2, & \text{if } M = 1 \\ 0, & \text{if } M = 0 \end{cases} \tag{6}$$

Here, M denotes the mask that identifies the known regions of the image. M takes the value of 1 in known regions and 0 elsewhere. ϵ represents the true noise, i.e., the result of adding noise to the original input image. ϵ_θ denotes the noise predicted by the model, and x_t is the image after being corrupted by noise at time step t.

4 Experiments

4.1 Datasets

We conduct experiments on two public UAV-view datasets: SUES-200 [23] and University-1652 [24]. SUES-200 contains drone images captured at multiple altitudes while circling buildings, with 50 frames sampled per scene across various environments such as parks, lakes, and residential areas. University-1652 simulates a spiral descent trajectory, and we estimate camera parameters using COLMAP [25].One-fifth of the UAV-view images in each dataset is used for testing.

4.2 Experimental Settings and Evaluation Metrics

All methods, including Crfill [15], BLD [17], MAT [16], SPA-Former [18], and our View-Diff, are implemented in PyTorch 2.0.1 with Python 3.8. Experiments are conducted on a single NVIDIA RTX 3090 GPU under Ubuntu 20.04. View-Diff is fine-tuned from a pre-trained diffusion model (v1.4) for 100 iterations using the Adam optimizer(learning rate 1×10^{-5}), 50 sampling steps.

To assess the quality of generated novel-view images, we use five metrics comparing them to ground truth. SSIM evaluates structural similarity (luminance, contrast, texture), while PSNR measures pixel-level distortion. LPIPS assesses perceptual similarity using high-level features, and I2I quantifies semantic alignment using CLIP embeddings. FID compares the feature distributions of real and generated images to reflect realism. These metrics jointly capture structural fidelity, perceptual quality, and semantic consistency.

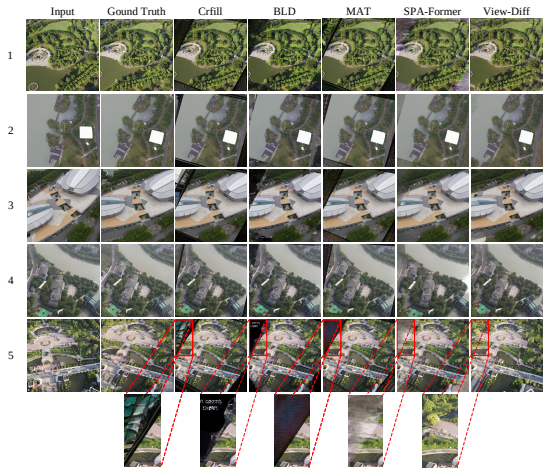

Fig. 4. Qualitative comparison of results from different methods for generating novel views.

4.3 Comparisons

Experiments are conducted on the SUES-200 and University-1652 datasets to quantitatively evaluate the quality of novel view generation. Since existing methods lack explicit viewpoint control, we preprocess the known regions to roughly align with the target view, while the unknown regions are generated by the comparative models.

Quantitative analysis. We present a quantitative comparison in Table 1 using standard metrics: SSIM, PSNR, LPIPS, I2I, and FID.

Table 1. Quantitative comparison results on SUES-200 and University-1652 datasets. ↑ indicates higher is better, ↓ indicates lower is better. Bold indicates the best value, and underline indicates the second-best value.

Dataset	Method	SSIM↑	PSNR↑	LPIPS↓	I2I↑	FID↓
SUES-200	Crfill [15]	0.3158	13.7536	0.3362	<u>0.8366</u>	498.9
	BLD [17]	0.2887	12.9124	0.3258	0.8158	457.1
	MAT [16]	<u>0.3169</u>	13.5061	0.3303	0.8164	561.4
	SPA-Former [18]	**0.3776**	**17.0964**	<u>0.3096</u>	0.8258	<u>338.1</u>
	View-Diff(ours)	0.3147	<u>16.0963</u>	**0.2873**	**0.8868**	**293.6**
University-1652	Crfill [15]	0.2553	12.7229	0.3785	0.8199	219.7
	BLD [17]	0.2387	11.9897	0.3758	0.8148	208.2
	MAT [16]	<u>0.2562</u>	12.4988	<u>0.3671</u>	0.7909	197.1
	SPA-Former [18]	**0.2986**	**14.2037**	0.3828	<u>0.8622</u>	<u>123.1</u>
	View-Diff(ours)	0.2474	<u>13.4524</u>	**0.3526**	**0.8631**	**108.7**

View-Diff consistently outperforms prior methods, especially in perceptual quality and semantic alignment. Unlike models that favor either local detail or global structure, View-Diff strikes a better balance through its Dual-Branch Pipeline and Optimized Perceptual Module. SPA-Former achieves high PSNR and SSIM but suffers from high LPIPS, indicating weak perceptual alignment. MAT restores local structures well but lacks global consistency (low I2I). View-Diff slightly compromises SSIM to enhance semantic coherence, which is more vital for novel view generation. Compared with Crfill and BLD, View-Diff delivers more robust results. Crfill's high FID shows poor realism; BLD's low LPIPS and I2I reveal weak semantic and spatial reasoning. View-Diff outperforms both in all key metrics.

Given the absence of standard measures for viewpoint controllability, we use LPIPS as a proxy. Its sensitivity to spatial and structural differences makes it suitable for evaluating viewpoint alignment. As shown in Table 1, View-Diff achieves the lowest LPIPS, confirming its structural consistency.

Qualitative analysis. As shown in Fig. 4, View-Diff consistently generates semantically coherent and structurally realistic novel views across diverse scenes, validating its effectiveness in explicit viewpoint conditioning and spatially guided generation. In Scene 1, Crfill and BLD produce artifacts, and MAT misrepresents vegetation, while View-Diff reconstructs natural grass with smooth transitions. In Scene 2, only View-Diff restores the road layout with accurate geometry and consistent textures. In Scene 3, MAT sharpens edges but fails to maintain global layout; View-Diff preserves alignment across rooftops and walls. In Scene 4, View-Diff delivers the most plausible spatial composition despite large missing regions. In Scene 5, SPA-Former struggles with structural complexity, but View-Diff retains fine details like road–grass boundaries.

Unlike prior methods that suffer from fragmented regions or hallucinated geometry. These issues are often caused by limited structural modeling or weak

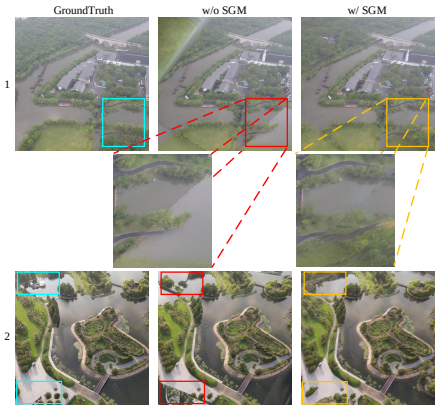

Fig. 5. Qualitative comparison of results with and without Spatial Guidance Module(SGM)

global priors. View-Diff uses a dual-branch pipeline to separate known and unknown regions. This ensures high-fidelity details in known regions. It also enables flexible and semantically aligned generation in unknown regions. With explicit viewpoint guidance, View-Diff produces globally coherent results.

Table 2. Comparison of LPIPS↓ with and without Spatial Guidance Module(SGM). Bold indicates the best value.

Category	W/ SGM	W/O SGM
Park	**0.2482**	0.2851
Lake	**0.2751**	0.3150
School	**0.2855**	0.3227
Residential Area	**0.3136**	0.3487

4.4 Ablation Experiments

To validate the effectiveness of View-Diff, we conduct two ablation studies on the Spatial Guidance Module and Optimized Perceptual Module, assessing their impact on structural consistency and perceptual quality in novel view generation.

Spatial Guidance Module. To demonstrate the effectiveness of the Spatial Guidance Module, we conducted experiments on the SUES-200 dataset, which contains more diverse scenes than University-1652. Images were grouped into four scene categories—parks, lakes, schools, and residential areas—with 38, 60,

17, and 85 images, respectively. To avoid explicit scene bias, no scene labels or prompts were used during inference, all images were processed uniformly, and results were categorized later for metric analysis. This setup better reflects generalization to unseen scenes. Due to its sensitivity to perceptual and structural

Table 3. Quantitative comparison results on the SUES-200 and University-1652 datasets using Optimized Perceptual Module(OPM) with different iteration counts. ↑ indicates higher is better, and the worst performance is indicated by dashed lines.

Dataset	Iteration Count	SSIM↑	PSNR↑	LPIPS↓	I2I↑	FID↓
SUES-200	0	0.3107	15.8002	0.3042	0.8490	314.1
	25	0.3140	15.8715	0.3093	0.8748	300.6
	50	0.3129	15.9072	0.3061	0.8833	291.7
	75	0.3140	16.0093	0.3021	0.8883	291.8
	100	0.3147	16.0963	**0.2873**	0.8868	293.6
	125	0.3153	16.1466	0.2972	**0.8922**	**290.7**
	150	**0.3173**	**16.2135**	0.2968	0.8915	293.6
University-1652	0	0.2487	13.3348	0.3665	0.8265	126.2
	25	0.2469	13.3439	0.3747	0.8572	120.9
	50	0.2467	13.3678	0.3715	0.8655	117.7
	75	0.2407	13.4188	0.3689	0.8689	112.5
	100	0.2474	13.4524	**0.3526**	0.8631	**108.7**
	125	0.2479	13.4993	0.3662	0.8709	109.5
	150	**0.2491**	**13.5473**	0.3657	**0.8709**	109.3

differences, LPIPS was used to evaluate spatial consistency between generated and ground-truth images. As shown in Table 2, incorporating the Spatial Guidance Module (SGM) consistently lowers LPIPS across all scenes, demonstrating improved robustness and spatial consistency.

Fig. 5 qualitatively compares results with and without SGM, highlighting significant improvements. In Scene 1, SGM maintains waterway continuity and clear separation from grassland, correcting discontinuities. In Scene 2, SGM restores smooth, aligned road geometry, resolving baseline fragmentation. These results confirm SGM's effectiveness in accurately modeling spatial structure.

Optimized Perceptual Module. To demonstrate the effectiveness of the Optimized Perceptual Module (OPM), Fig. 6 shows results at different optimization iterations. Without optimization (0 iterations), the model overfits local textures and neglects global context, causing artifacts and structural inconsistencies. Semantic understanding is poor, leading to errors in unknown regions—for example, unclear roads in Scene 1, structural deviations in Scene 2, and misreplacement of a bridge in Scene 3. After 100 iterations, these issues are greatly reduced: the model better captures scene semantics and accurately reconstructs key elements like grasslands and bridges. This shows that OPM enhances un-

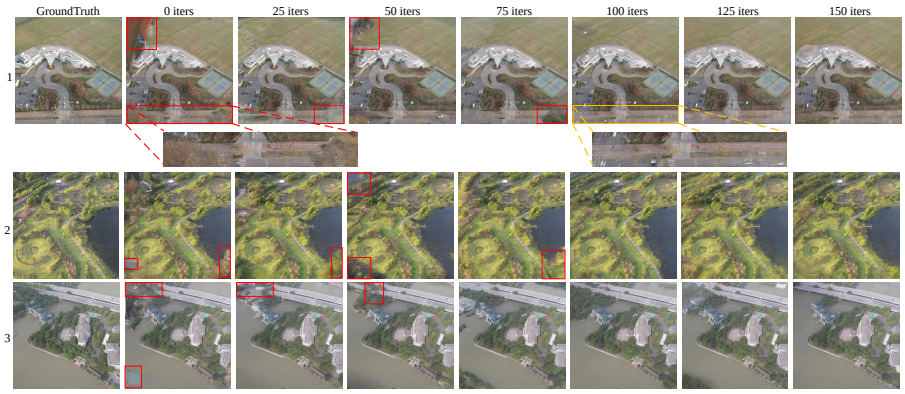

Fig. 6. Qualitative comparison of results with and without Optimized Perceptual Module (OPM).

derstanding of known regions, enabling coherent global semantic integration and higher-quality, consistent generation. Quantitative results in Table 3 confirm that performance improves with more optimization iterations, demonstrating OPM's effectiveness and stability in preserving features and enhancing perceptual quality.

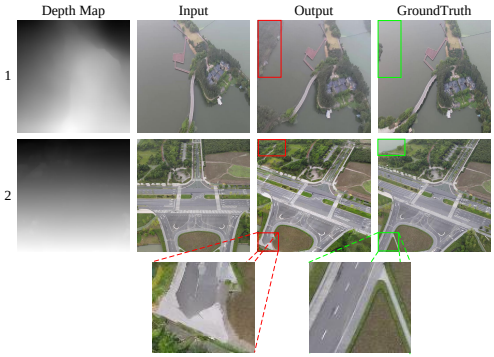

Fig. 7. View generation results with depth maps. Red boxes highlight distortions caused by depth errors.

5 Conclusion and Future Work

To tackle geometric inconsistency, texture discontinuity, and imprecise viewpoint control in aerial novel view generation, we propose *View-Diff*, a controllable

diffusion-based framework. It introduces a Dual-Branch Pipeline that separates viewpoint transformation from spatial completion, improving handling of known and unknown regions. An Optimized Perceptual Module further enhances semantic understanding and global structure in known areas, guiding realistic generation in unknown regions. Experiments on public aerial datasets show *View-Diff* outperforms baselines, confirming its effectiveness.

Although View-Diff shows strong performance in multi-view generation from single-view inputs, it depends on monocular depth estimation, where minor errors can impact geometric accuracy in complex scenes. As shown in Fig. 7, Scene 1 reveals unclear depth transitions between land and water, causing shoreline distortion and unnatural deformation. In Scene 2, oversmoothed depth in vegetation leads to misaligned geometry and warped roads. These issues highlight how depth inaccuracies can degrade structural fidelity. Future work will aim to reduce reliance on depth estimation and explore complementary strategies for improved spatial perception.

Acknowledgments. This work was supported by the National Key R&D Program of China (2022YFF0604502).

References

1. Yanhao Chu, Qiang Tong, Xuhong Liu, and Xiulei Liu. Odadapter: An effective method of semi-supervised object detection for aerial images. In *Chinese Conference on Pattern Recognition and Computer Vision (PRCV)*, pages 158–172. Springer, 2024.
2. Rundi Wu, Ben Mildenhall, Philipp Henzler, Keunhong Park, Ruiqi Gao, Daniel Watson, Pratul P Srinivasan, Dor Verbin, Jonathan T Barron, Ben Poole, et al. Reconfusion: 3d reconstruction with diffusion priors. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 21551–21561, 2024.
3. Diederik P. Kingma and Max Welling. Auto-encoding variational bayes. In *Proceedings of the 2nd International Conference on Learning Representations (ICLR)*, Banff, Canada, 2014.
4. Eunbyung Park, Jimei Yang, Ersin Yumer, Duygu Ceylan, and Alexander C Berg. Transformation-grounded image generation network for novel 3d view synthesis. In *Proceedings of the ieee conference on computer vision and pattern recognition*, pages 3500–3509, 2017.
5. Ian J Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. Generative adversarial nets. *Advances in neural information processing systems*, 27, 2014.
6. Yu Tian, Xi Peng, Long Zhao, Shaoting Zhang, and Dimitris N Metaxas. Crgan: learning complete representations for multi-view generation. *arXiv preprint arXiv:1806.11191*, 2018.
7. Xiaogang Xu, Ying-Cong Chen, and Jiaya Jia. View independent generative adversarial network for novel view synthesis. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 7791–7800, 2019.
8. Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *Advances in neural information processing systems*, 33:6840–6851, 2020.

9. Eric R Chan, Koki Nagano, Matthew A Chan, Alexander W Bergman, Jeong Joon Park, Axel Levy, Miika Aittala, Shalini De Mello, Tero Karras, and Gordon Wetzstein. Generative novel view synthesis with 3d-aware diffusion models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 4217–4229, 2023.
10. XL Tang, YM Du, YW Liu, JX Li, and YW Ma. Image recognition with conditional deep convolutional generative adversarial networks. *Acta Automatica Sinica*, 44(5):855–864, 2018.
11. Jun-Yan Zhu, Taesung Park, Phillip Isola, and Alexei A Efros. Unpaired image-to-image translation using cycle-consistent adversarial networks. In *Proceedings of the IEEE international conference on computer vision*, pages 2223–2232, 2017.
12. Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. *arXiv preprint arXiv:2010.02502*, 2020.
13. Christopher B Choy, Danfei Xu, JunYoung Gwak, Kevin Chen, and Silvio Savarese. 3d-r2n2: A unified approach for single and multi-view 3d object reconstruction. In *Computer vision–ECCV 2016: 14th European conference, amsterdam, the netherlands, October 11-14, 2016, proceedings, part VIII 14*, pages 628–644. Springer, 2016.
14. Hao Tang, Dan Xu, Nicu Sebe, Yanzhi Wang, Jason J Corso, and Yan Yan. Multi-channel attention selection gan with cascaded semantic guidance for cross-view image translation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 2417–2426, 2019.
15. Yu Zeng, Zhe Lin, Huchuan Lu, and Vishal M Patel. Cr-fill: Generative image inpainting with auxiliary contextual reconstruction. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 14164–14173, 2021.
16. Wenbo Li, Zhe Lin, Kun Zhou, Lu Qi, Yi Wang, and Jiaya Jia. Mat: Mask-aware transformer for large hole image inpainting. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10758–10768, 2022.
17. Omri Avrahami, Ohad Fried, and Dani Lischinski. Blended latent diffusion. *ACM transactions on graphics (TOG)*, 42(4):1–11, 2023.
18. Wenli Huang, Ye Deng, Siqi Hui, Yang Wu, Sanping Zhou, and Jinjun Wang. Sparse self-attention transformer for image inpainting. *Pattern Recognition*, 145:109897, 2024.
19. René Ranftl, Katrin Lasinger, David Hafner, Konrad Schindler, and Vladlen Koltun. Towards robust monocular depth estimation: Mixing datasets for zero-shot cross-dataset transfer. *IEEE transactions on pattern analysis and machine intelligence*, 44(3):1623–1637, 2020.
20. Christoph Schuhmann, Romain Beaumont, Richard Vencu, Cade Gordon, Ross Wightman, Mehdi Cherti, Theo Coombes, Aarush Katta, Clayton Mullis, Mitchell Wortsman, et al. Laion-5b: An open large-scale dataset for training next generation image-text models. *Advances in neural information processing systems*, 35:25278–25294, 2022.
21. Meng-Li Shih, Shih-Yang Su, Johannes Kopf, and Jia-Bin Huang. 3d photography using context-aware layered depth inpainting. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 8028–8038, 2020.
22. Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. In *Medical image computing and computer-assisted intervention–MICCAI 2015: 18th international conference, Munich, Germany, October 5-9, 2015, proceedings, part III 18*, pages 234–241. Springer, 2015.

23. Runzhe Zhu, Ling Yin, Mingze Yang, Fei Wu, Yuncheng Yang, and Wenbo Hu. Sues-200: A multi-height multi-scene cross-view image benchmark across drone and satellite. *IEEE Transactions on Circuits and Systems for Video Technology*, 33(9):4825–4839, 2023.
24. Zhedong Zheng, Yunchao Wei, and Yi Yang. University-1652: A multi-view multi-source benchmark for drone-based geo-localization. In *Proceedings of the 28th ACM international conference on Multimedia*, pages 1395–1403, 2020.
25. Johannes L Schonberger and Jan-Michael Frahm. Structure-from-motion revisited. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 4104–4113, 2016.